Venkata Lakshmi Parimala Pasupuleti

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github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

AI Engineer with 9 years of production Python and full-stack engineering, specializing in **AI agent systems, RESTful API integration, and enterprise-scale service delivery**. Built AgentProbe -- an open-source multi-agent AI toolkit (LangGraph, FastAPI, ChromaDB) with CI/CD quality gates and LLM-as-judge evaluation. Delivered **HIPAA-compliant AI on structured healthcare records** (AI on FHIR) and production AI pipelines across regulated financial data. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 -- Aug 2025

Research Associate -- Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; designed Tableau, ggplot2, and Quarto dashboards improving faculty research comprehension by 40%.
- Applied Isolation Forest to automate data-quality checks across 3 research pipelines -- surfaced anomalies manual review missed, improving downstream model reliability.

Entrata
Jan 2017 -- Aug 2024

Software Engineer

- Designed and maintained **20+ RESTful API-driven microservices** (Python, PostgreSQL, RabbitMQ, FastAPI) across 120+ Agile sprints for 100K+ monthly active users -- 40% deployment time reduction; 99.5% uptime over 6+ months.
- Applied Python, PyTorch, and OpenAI API to build **AI pipelines across 1.6M financial records** -- production AI on regulated data with output validation, quality gates, and compliance controls; cut microservices execution time 50% via Docker containerization.
- Achieved **100% PHI compliance** migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; built affordable housing compliance modules (Section 8/HCV, LIHTC, HUD) - income eligibility verification, AMI threshold calculations, and HUD regulatory reporting across 100K+ unit portfolios; improved query performance 30% tuning 200+ SQL queries; led 300+ peer code reviews across all engineering teams.

PROJECTS

AgentProbe: Multi-Agent AI Testing Toolkit
Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider model routing -- deployed as a **15-endpoint RESTful FastAPI backend** with Docker and CI/CD integration.
- Implemented LLM-as-judge evaluation engine (factual accuracy, hallucination, retrieval, completeness, compliance) with automated quality gates -- failing agent runs blocked before production; React dashboard tracking per-run metrics and system health.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% building a **REST API-integrated NLP platform** translating natural language queries (EN/ES) to FHIR healthcare APIs using LLMs, BERT, and semantic retrieval -- **HIPAA-compliant RBAC** across structured medical records.

PropLens: Lease Intelligence & Compliance AI
In Progress

Python · LangGraph · ChromaDB · FastAPI · LLMs · React

- Building agent-based RAG pipeline over lease PDFs -- LLM-as-judge scoring and anomaly detection across 1,000+ tenant records; RESTful FastAPI backend with compliance gap detection and automated reporting.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University
Graduated May 2026
GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Jawaharlal Nehru Technological University, Anantapur
2013 -- 2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Systems, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Prompt Engineering, Agent Observability, ChromaDB, Embeddings, OpenAI API, Anthropic API, Hugging Face, FastAPI, RESTful APIs

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, BERT, spaCy, NumPy, Pandas, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL, Oracle